

Yuchen Wang, PhD

Postdoctoral Associate

Simons Center for Computational Physical Chemistry

New York University

Tel: 785-317-1859

Email: yw8889@nyu.edu

Education and Experience

- Postdoctoral Associate, Department of Chemistry at New York University, NY, USA
Major Professor: Prof. Norah M. Hoffmann 2024 - present
- Ph.D. in Chemistry, Kansas State University, Manhattan, KS, USA
Major Professor: Prof. Christine M. Aikens 2019 - 2024
- B.Sc. in Chemistry, Anhui Normal University, Anhui, China 2015 - 2019

Research Areas

- Theory and simulations for plasmonic nanoparticles
- Time-dependent density functional theory (TDDFT)
- Semi-empirical methods development
- Machine-learning potential development
- Non-adiabatic molecular dynamics (NAMD) of photo-induced process
- Polariton chemistry and cavity-quantum electrodynamics (cQED)

Software Development

- ADF (TDDFT-aas method energy and analytical gradient implementation)
- Libra (Implementation of nonadiabatic dynamics with polaritonic chemistry effects)
- PySCF (Implementation of real-time TDDFT for polaritonic chemistry)

Technical Skills

- Quantum chemistry: Gaussian; GAMESS-US; ORCA; ADF; CP2K
- Plane wave based: Quantum Espresso; VASP; GPAW
- Molecular dynamics: LAMMPS; Libra
- Coding language: C++; Fortran; Python

Selected Awards

1. Graduate Student Research Award at Kansas State University April 2024
2. Scott Fateley Memorial Award at Kansas State University April 2023

3. Chemistry Alumni Graduate Student Award at Kansas State University

August 2022

Conference Presentations (Oral)

1. Oral presentation: Inducing long exciton lifetimes: Laser versus Voltage driven excitations in STM-break junction. Molecular Quantum Information Seminar Series, University of Minnesota, June 2025
2. Oral presentation: Inducing long exciton lifetimes: Laser versus Voltage driven excitations in STM-break junction. New York Polaritonic Meeting, NYU-CCQ, May 2025
3. Oral presentation: Inducing long exciton lifetimes: Laser versus Voltage driven excitations in STM-break junction. Texas Computational Symposium, Feb 2025
4. Oral presentation: Advanced Computational Methods for Studying Noble Metal Nanoclusters: From Quantum Chemistry to Machine Learning. Hunan Agricultural University, Dec 2024
5. Oral presentation: Implementation of energy and gradient for TDDFT-approximate auxiliary function (aas) method. Wang, Yuchen and Aikens, Christine M. 2024, Mar. 15 VISTA talk, virtual
6. Oral presentation: Implementation of energy and gradient for TDDFT-approximate auxiliary function (aas) method. Wang, Yuchen and Aikens, Christine M. 2023, Nov. 15 - Nov. 18 ACS Southwest regional meeting, Oklahoma City, OK
7. Oral presentation: Static field and continuous wave field effects on plasmon-induced H₂ activation. Wang, Yuchen and Aikens, Christine M. ACS Fall 2023, Aug 13 - Aug 17, 2023, San Francisco, CA
8. Lightning talk: Active learning applied to constructing force fields for gold thiolate-protected nanoclusters. Wang, Yuchen and Aikens, Christine M. 4th Artificial Intelligence for Materials Science (AIMS) workshop. July 25 - July 27, 2023, Virtual

Teaching Experience

- Mandarin instructor in International Student Center at K-State 2022 - 2023
- Graduate teaching assistant for Chemistry I and II laboratories 2019 - 2022
- Chemistry help room teaching assistant 2019 - 2021
- Mentoring undergraduate students in Prof. Aikens lab:
Jacqueline Pinkerton (REU program); Aidan Lindsay (REU program); Brie Little
- Mentoring students in Prof. Hoffmann lab:
Oliver Tan (graduate student); Jihye Han (rotation graduate student); Mahika Ralhan (high school student)

Services

- Journal reviewer: JACS Au; Inorganic chemistry; Digital discovery

- Research mentor: REU program; GSTEM program at NYU Courant

Publications

In preparation:

1. Y. Wang #, O. Tan#, and N. Hoffmann Inducing long exciton lifetimes: Laser versus voltage driven excitations and strong light-matter interaction in STM-break junction. In preparation.
2. Y. Wang, M. Shakiba, V. A. Akimov, and N. Hoffmann, Implementation and benchmarking decoherence corrections in polaritonic non-adiabatic molecular dynamics. In preparation.

Peer reviewed:

1. Wang, Y., Samarasinghe, D. S. N. D., Deng, H., Liu, B. and Aikens, C. M. Gaussian process approach to constructing transferable force fields for gold thiolate-protected nanoclusters. *J. Chem. Inf. Model.* 2025, 65 (8), 3892-3902 (Selected as the front cover)
2. Sun, X#, Wang, Y#., Wu, Q#, Han, Y.Z., Gong, X., Tang, X., Aikens, C.M., Shen, H. and Zheng, N., 2025. Cu₆₆ nanoclusters from hierarchical square motifs: Synthesis, assembly, and catalysis. *Aggregate*, 6(1), p.e651.
3. Wang, Z.#, Wang, Y.#., Zhang, C., Zhu, Y.J., Song, K.P., Aikens, C.M., Tung, C.H. and Sun, D., 2024. Silvery fullerene in Ag₁₀₂ nanosaucer. *National Science Review*, 11(7), p.nwae192.
4. Wang, Y., Havenridge, S., Aikens, C. M. Implementation of energy and gradient for the TDDFT- approximate auxiliary function (aas) method. *J. Chem. Phys.* 2024, 161, 024101
5. Wang, Z., Wang, Y., Li, L., Aikens, C. M., Gao, ZY., Tung, CH., Sun, D. Temperature-Driven Selective Formation of Silver Nanoclusters and Their Transformation to the Same Product Following Different Routes. *Angew. Chem. Int. Ed.* 2024, e202403464.
6. Wang, Y. & Aikens, C. M. Effects of nanowire doping on plasmon-enhanced N₂ dissociation. *Phys. Chem. A* 2024, 128, 19, 3784-3793
7. Truttmann, V.; Loxha, A.; Banu, R.; Pittenauer, E.; Malola, S.; Matus, M. F.; Wang, Y.; Ploetz, E. A.; Rupprechter, G.; Bürgi, T.; Häkkinen, H.; Aikens, C.; Barrabés, N. Directing Intrinsic Chirality in Gold Nanoclusters: Preferential Formation of Stable Enantiopure Clusters in High Yield and Experimentally Unveiling the “Super” Chirality of Au₁₄₄. *ACS Nano* 2023, 17 (20), 20376-20386.
8. Wang, Y. & Aikens, C. M. Connectivity between Static Field and Continuous Wave Field Effects on Excitation-Induced H₂ Activation. *J. Phys. Chem. C* 2023, 127, 31, 15375-15384
9. Wang, Y. & Aikens, C. M. Effects of Field Strength and Silver Nanowire Size on Plasmon Enhanced N₂ Dissociation *J. Phys. Chem. A* 2023, 127, 27, 5609-5619
10. Wang, Y., Li, A., Pinkerton, J., & Aikens, C. M. Effects of Diglyme on Au Nanocluster Formation: Mechanism, 1H NMR, and Bonding. *J. Phys. Chem. A* 2022, 126, 42, 7598-7605
11. Anderson, I. D., Wang, Y., Aikens, C. M., & Ackerson, C. J. An ultrastable thiolate/diglyme ligated cluster: Au₂₀(PET)₁₅(DG)₂. *Nanoscale*, 14(25), 9134-9141.